

Monetary Policy Uncertainty and the Dispersion of Inflation Expectations in South Africa

Research Proposal

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Motivation

Since the Global Financial Crisis, uncertainty has re-emerged as a first-order driver of macroeconomic outcomes, affecting investment, hiring, and consumption decisions (Bloom 2009). A growing strand of this literature focuses specifically on *monetary policy uncertainty* (MPU) – uncertainty about the future stance and reaction function of central banks – and its effect on the formation of inflation expectations (Arce-Alfaro and Blagov 2023). Where the private sector is less certain about the monetary authority’s future actions, it should also be less certain, and more disperse, in its inflation forecasts. For a small, open, inflation-targeting economy such as South Africa, this channel is compounded by exchange-rate uncertainty and spillovers from US monetary policy uncertainty, both of which have been shown to matter for domestic volatility and expectations formation (Mumtaz and Zanetti 2013; Bianchi, Kung, and Tirsikh 2023).

Despite this, little empirical work has examined how MPU – domestic and foreign – relates to the *cross-sectional dispersion* of inflation expectations across different types of economic agents in South Africa. Disagreement (dispersion) is itself informative: it proxies the confidence agents place in their own forecasts and, unlike point forecasts, captures how uncertainty is transmitted unevenly across the economy.

Research Questions

1. Does monetary policy uncertainty – in South Africa and in the United States – raise the dispersion of inflation expectations among South African businesses, financial analysts, and labour representatives?

2. Does exchange-rate uncertainty (USD/ZAR volatility) amplify or operate independently of this relationship?
3. Does the sensitivity of expectations dispersion to MPU vary by forecast horizon (current year through five years ahead) or by respondent sector?

Data

The analysis draws on quarterly data assembled from the following sources:

Series	Source	Frequency
Dispersion (SD) of inflation expectations, by sector (business, finance, labour) and horizon	BER Inflation Expectations Survey	Quarterly
South African monetary policy uncertainty	Own GARCH(1,1) conditional volatility, policy rate	Monthly, aggregated
Alternative South African monetary policy uncertainty	Codera news based measure	Monthly, aggregated
US monetary policy uncertainty	Baker, Bloom & Davis EPU – monetary policy category	Monthly, aggregated
USD/ZAR exchange-rate uncertainty	Own GARCH(1,1) conditional volatility, USD/ZAR daily returns	Daily, aggregated

All series are harmonised to a common quarterly frequency, spanning approximately 1985 (uncertainty series) to 2026, with the dispersion measures available from 2011 onward.

Proposed Methodology

1. **Uncertainty measurement:** estimate a GARCH(1,1) model on USD/ZAR log returns to derive a time-varying, model-based exchange-rate uncertainty proxy. A similar approach for the MPU.
2. **Empirical model:** regress each sector-horizon dispersion measure on lagged domestic and US MPU, exchange-rate volatility, and relevant macro controls (e.g. inflation, policy rate, output gap), using HAC-robust standard errors to address serial correlation typical of overlapping quarterly survey data. A Bayesian approach is possible given some gaps in the responses of the BER survey.
3. **Robustness:** assess sensitivity to horizon and sector, and to alternative uncertainty proxies (news based measures); check series for stationarity and structural breaks given the long sample period (repo-era transition, 2008 GFC, 2020 pandemic).

Contribution

This project would provide the first South African evidence linking monetary policy uncertainty – both domestic and US-sourced – to disaggregated inflation-expectations dispersion, combining a model-based FX uncertainty measure with survey-based MPU and disagreement indices in a single, harmonised quarterly dataset.

References

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